Eprueba 2.

$$X = \begin{bmatrix} 0.5 \\ 0.8 \end{bmatrix} \qquad y = 1$$

$$W_1 \begin{bmatrix} 0.1 & 0.4 \\ 0.2 & 0.5 \\ 0.3 & 0.6 \end{bmatrix} \qquad b_1 = \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 0.2, 0.3, 0.4 \end{bmatrix} \qquad b_2 = \begin{bmatrix} 0.1 \end{bmatrix}$$

Forward Pass.

$Z_1 = W_1 x + b_1 \qquad a_1 = \mathrm{ReLU}(z_1)$

$Z_1 = 0.1 \cdot 0.5 + 0.4 \cdot 0.8 + 0.1 = 0.47.$

$Z_2 = 0.2 \cdot 0.5 + 0.5 \cdot 0.8 + 0.2 = 0.70.$

$Z_3 = 0.3 \cdot 0.5 + 0.6 \cdot 0.8 + 0.3 = 0.93$

$$\mathrm{ReLU} = \max(0, z_1) = \begin{pmatrix} 0.47 \\ 0.70 \\ 0.93 \end{pmatrix}.$$

$$Z^{2} = W_2 h^{(1)} + b_2$$

$$Z^2 = 0.8 \cdot 0.47 + 0.3 \cdot 0.70 + 0.4 \cdot 0.93 \cancel{+0.2} = \underline{0.77}$$

Συναρτηση Sigmoid ειναι $\sigma(z) = \dfrac{1}{1+e^{-z}}$

Αρα $A = \sigma(0.77) = \dfrac{1}{1.46} \Rightarrow 0.68$

Απωλεια συναρτηση δυαδικης διεντροπιας Bce

$-\dfrac{1}{N} \log L(D)$ $N=1$ $y=1$

$$L(1,y) = -\left[1 \cdot \ln(0.68) + (1-1) \cdot \ln(1-0.68)\right]$$
$$= -\ln(0.68) = \boxed{0.37}$$

Gradient Descent

Επιπεδο εξοδου $\hat{y} = \sigma(z_2)$

$$\frac{\theta L}{\theta z} = \hat{y} - y.$$

Αρα.

$$\frac{\theta L}{\theta z} = 0.68 - 1 = -0.32.$$

Παραγωγος βαρ + bιαs.

$$\frac{\theta L}{\theta w_2} = \begin{pmatrix} -0.31 \end{pmatrix} \times \begin{pmatrix} 0.47 \\ 0.70 \\ 0.93 \end{pmatrix} \Rightarrow \begin{pmatrix} -0.222 \\ -0.140 \\ -0.293 \end{pmatrix}$$

$$\frac{\theta L}{\theta A_2} = \begin{pmatrix} -0.31 \end{pmatrix} \times \begin{pmatrix} 0.2 \\ 0.3 \\ 0.4 \end{pmatrix} \Rightarrow \begin{pmatrix} -0.06 \\ -0.09 \\ -0.12 \end{pmatrix}$$

Παραγωγος στο πρωτο επιπεδο

$$\frac{\theta L}{\theta w} = \begin{bmatrix} 0.06 \\ 0.09 \\ 0.12 \end{bmatrix} \times \begin{bmatrix} 0.5 & 0.8 \end{bmatrix} = \begin{bmatrix} -0.03 & -0.05 \\ -0.04 & 0.07 \\ -0.06 & 0.10 \end{bmatrix}$$

Ενημέρωση παραμέτρων

$\theta_{t+1} = \theta_t - n\,\theta_1(\theta_t)$

$b_{\text{νέο}} = 0.2 - 0.2 \times (-0.11) = 0.23$

$\theta_2 = \theta_{\cancel{...}} - n \frac{\theta\lambda}{\theta}$

$$W_2 = W_2 - 0.2 \cdot \frac{\partial L}{\partial W} = \begin{bmatrix} 0.2 & 0.4 \\ 0.2 & 0.5 \\ 0.3 & 0.6 \end{bmatrix}$$

$$- 0.2 \begin{bmatrix} -0.03 & -0.05 \\ -0.04 & -0.07 \\ -0.06 & 0.07 \\ & 0.10 \end{bmatrix} \Rightarrow \text{πράξεις.}$$

$$b_2 = \begin{bmatrix} 0.2 \\ 0.2 \\ 0.3 \end{bmatrix} - 0.2 \begin{bmatrix} -0.06 \\ -0.09 \\ -0.12 \end{bmatrix} \Rightarrow \text{πράξεις.}$$

~~$W_4 = [0.2\ 0.3\ 0.4] - 0.2 [\ldots]$~~